

# FrescoDiffusion

## 4K Image-to-Video with Prior-Regularized Tiled Diffusion

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### Task context

- Native-resolution I2V cannot process **complex 4K fresco-like images**.

- Current approaches and limitations :

|                       | Downsampled input to I2V | Tiled Diffusion |
|-----------------------|--------------------------|-----------------|
| Fine Details          | ✗                        | ✓               |
| Fidelity to the input | ✓                        | ✗               |

- FrescoDiffusion regularizes tiled predictions with a **global video prior** at every diffusion step in a **training-free** manner.

### FrescoArchive dataset

371 manually curated, complex multi-scene image-caption pairs

Steps (from LAION):

- Resolution filtering
- Fresco detection
- Deduplication + Captioning
- Manual curation

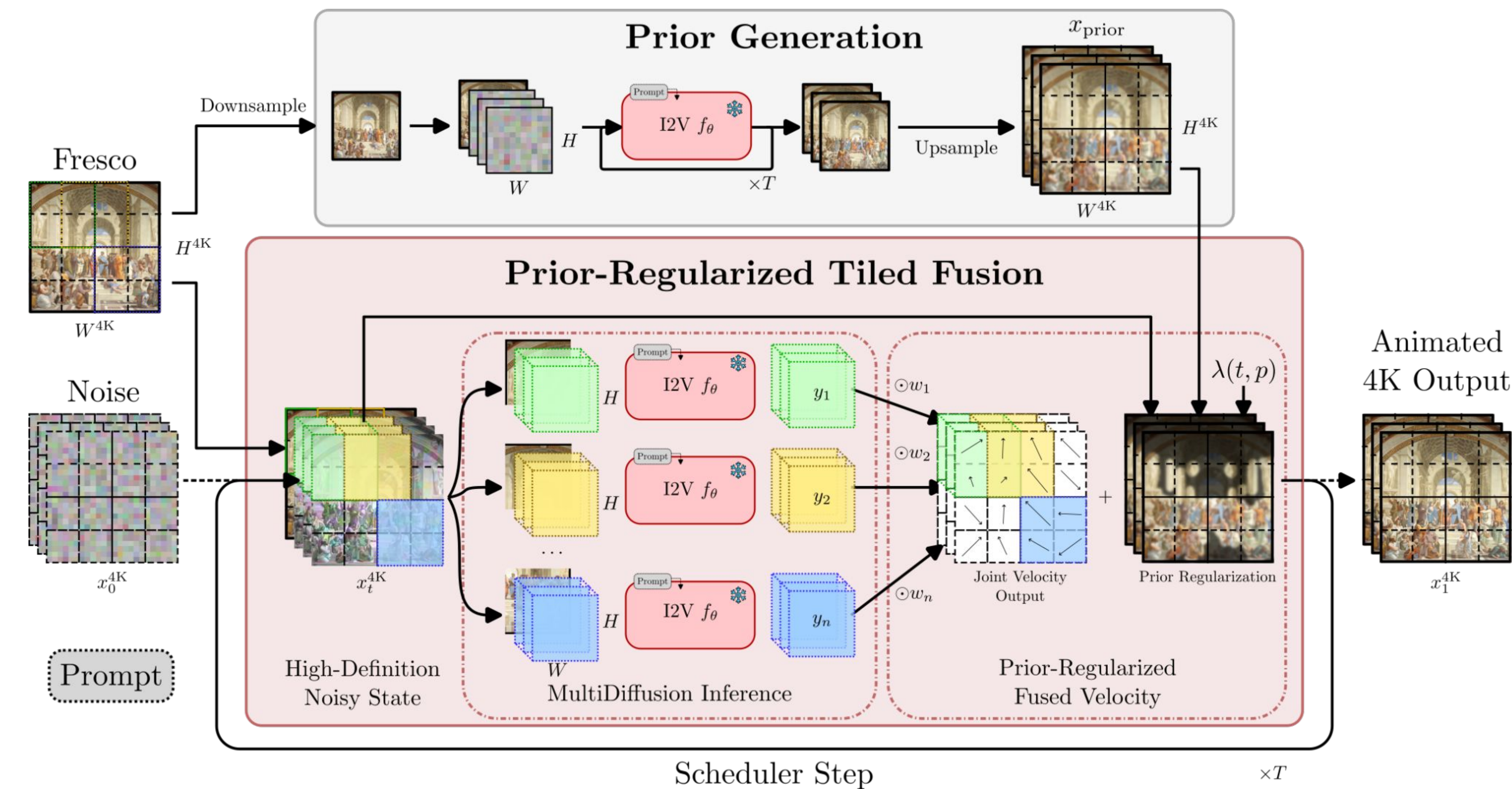


### Project page & video results

Additional videos, dataset, examples and resources on our website



### Methodology



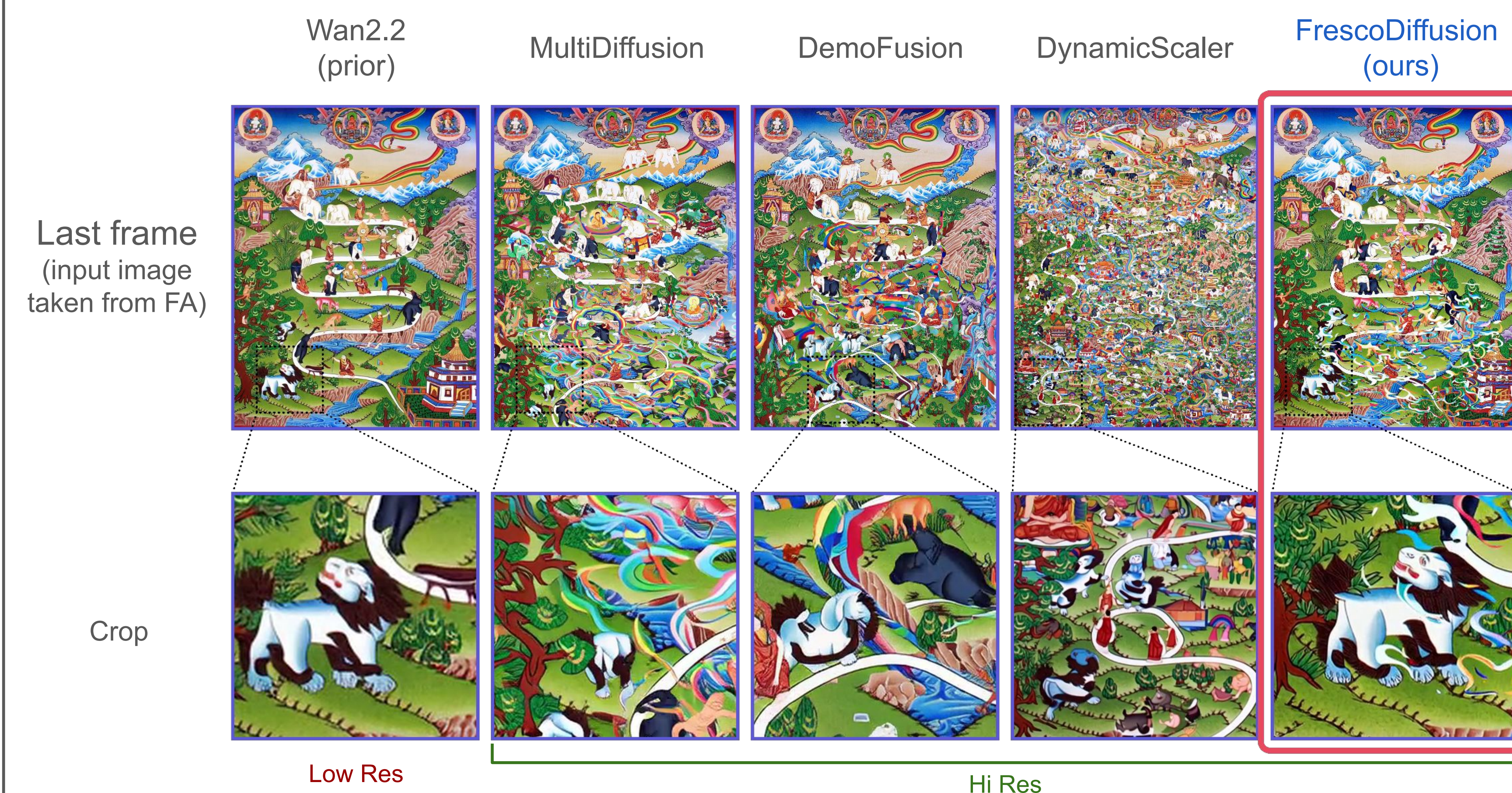
- A **two-step** generation process
- Generate the **low-resolution prior**.
  - Regularize** diffusion with the prior.

$$y_{FD}(x_t^{4K}) = \frac{\sigma_t \cdot \lambda \odot (x_t^{4K} - x_{prior}) + \sum_{i=1}^n w_i \odot y_i(x_t^{4K})}{\sigma_t^2 \cdot \lambda + \sum_{i=1}^n w_i}$$

$\lambda \rightarrow$  **controls similarity** between prior and generated image

- low  $\lambda$  permits creative local motion
- high  $\lambda$  preserves global structure.

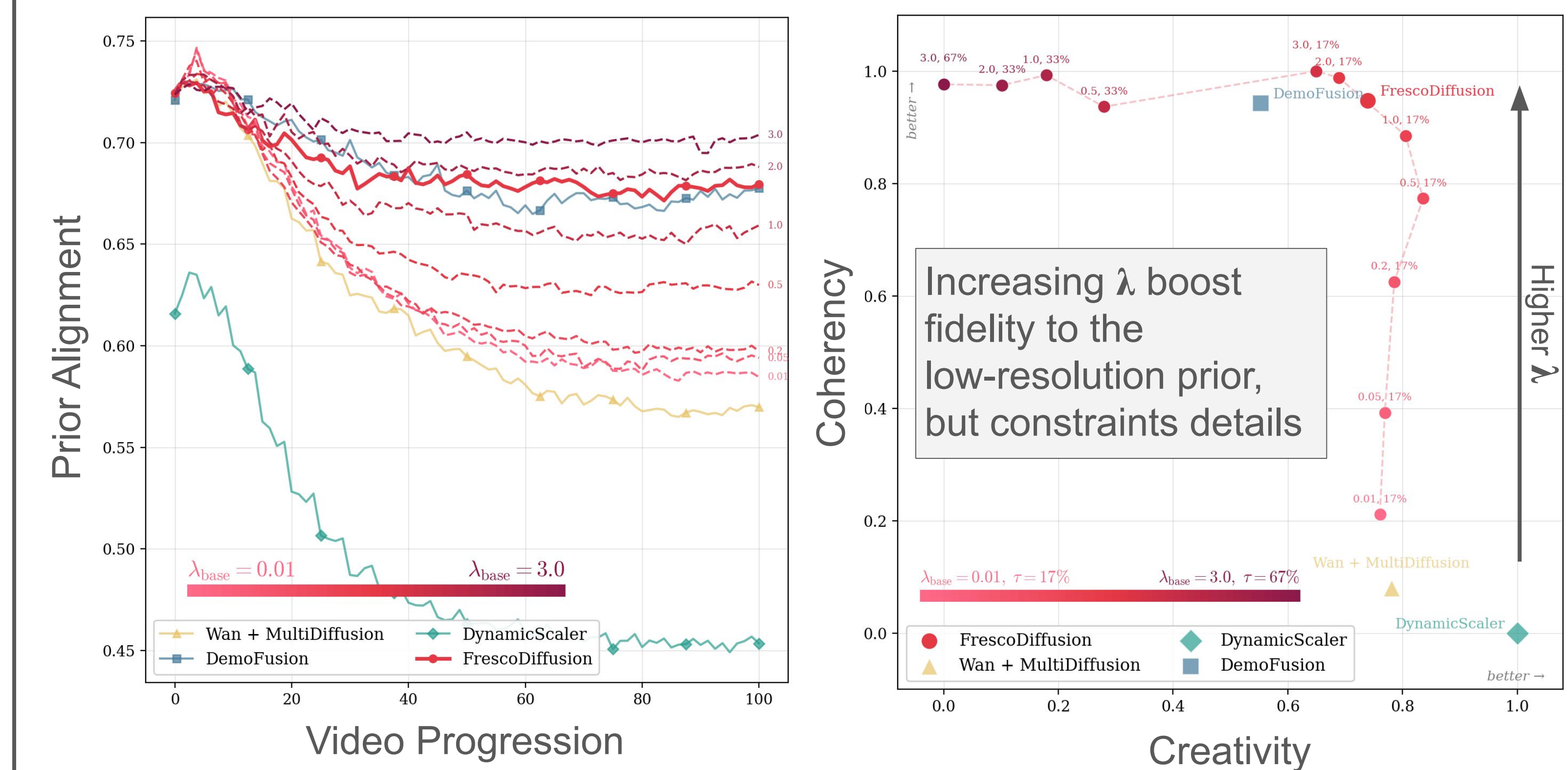
### Experimental Results



Our results are visually **preferred** (user study, 1465 ratings)

### Controlling FrescoDiffusion

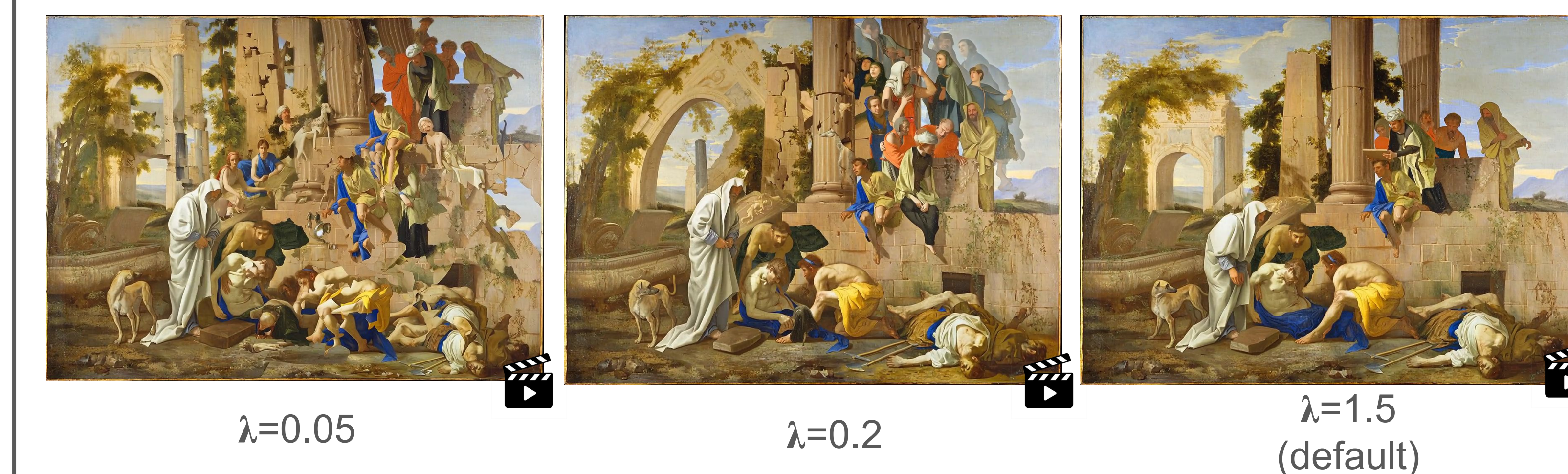
Coherency / creativity trade-off



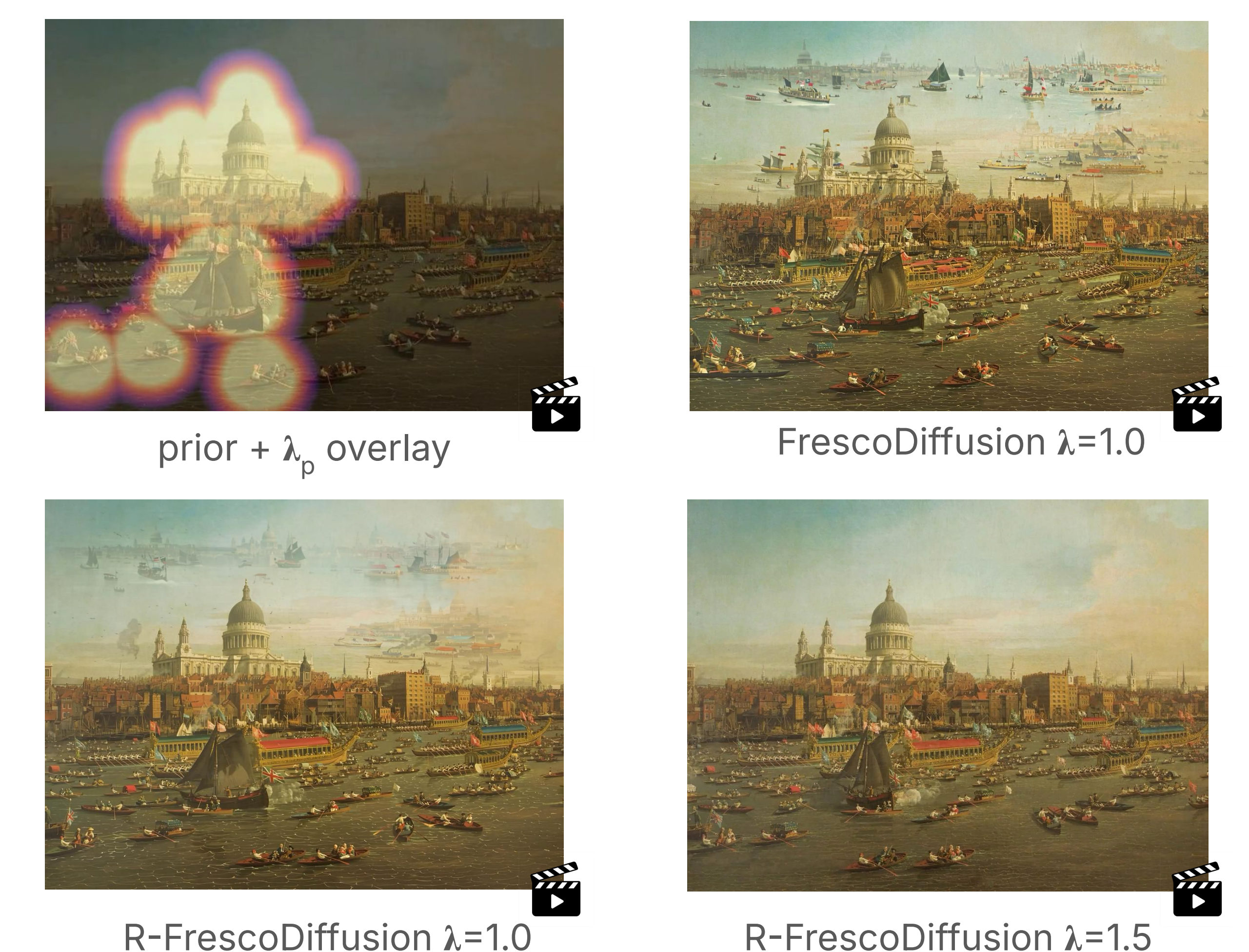
- Less similar
- More creative



- Highly similar
- Less creative



R-FrescoDiffusion (regional variant)



Controlling  $\lambda$  regionally enables animating differently each region.